



Yalamanchili Harsha Vardhan
Computer Science & Engineering
Indian Institute of Technology Bombay

24B1069
B.Tech.
Gender: Male
DOB: 03/01/2007

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2028	9 Core: 62 Total: 62

SCHOLASTIC ACHIEVEMENTS

- Attained **All India Rank 170** in the **Joint Entrance Examination Advanced**, among **180,000+** aspirants 2024
- Secured **Rank 84** in **TS EAPCET**, a premier state-level entrance exam, among **254,000+** contenders 2024
- Achieved **All India Rank 267** in the **Joint Entrance Examination Main**, among **1,400,000+** aspirants 2024
- Secured **Rank 119** in **AP EAPCET**, a major state-level entrance test, out of **339,000+** participants 2024

KEY PROJECTS

Summer Of Quant | *Quant Community, IIT Bombay* (June - July 2025)

- Studied **Markov chains**, **stochastic** and **Poisson processes** with applications in quantitative modeling
- Gained exposure to **time series analysis**, **feature engineering**, and signal generation for trading strategies
- Learned **quantitative trading strategy design**, focusing on risk management, **backtesting**, and model robustness
- Secured a **Certificate of Excellence** and ranked among the top 10 performers in the Summer of Quant program

Competitive Programming | *Summer Of Science* (June - July 2025)

Maths and Physics Club, IIT Bombay

- Explored recursion, **dynamic programming**, string operations, **greedy methods**, and **mathematical algorithms**
- Solved over **150+** Codeforces problems and puzzles across diverse difficulty levels to build strong analytical thinking
- Applied advanced data structures in coding challenges, emphasizing efficiency and modularity in implementation
- Worked with graph algorithms (**Dijkstra's**, **Floyd-Warshall**) and **sorting techniques** to optimize performance

Innovative Python Tools & Applications | *Self Projects* (June 2024 - Present)

- Engineered an automated **Sudoku solver** with **OpenCV**, using **image processing** and **backtracking** to handle noisy input efficiently, accurately interpreting grids and solving challenging Sudoku puzzles in real-time applications
- Developed a Windows utility to adjust the opacity of any running application for improved multitasking and focus
- Built a Python script to fetch and display all problems solved by a user on **Codeforces**, tracking progress efficiently
- Created a PDF converter that transforms documents into a dark-themed format, reducing the strain on eyes

Advanced Algorithm Implementation | *Course Project* (Aug - Sept 2025)

Guide: Prof. Ashutosh Gupta - Data Structures and Algorithms

- Implemented a **Red-Black Tree** with insert/delete operations and an optimized **hash table** using linear probing
- Devised **Strassen's matrix multiplication** with advanced memory optimizations, improving its efficiency
- Investigated and benchmarked **Heaps**, **Binary Trees**, and **BSTs** to evaluate efficiency and operational trade-offs

Restaurant E-Commerce Website | *Web Development Project* (January - February 2025)

Devcom, IIT Bombay

- Developed a full-stack **restaurant web application** featuring an **interactive menu interface** and a **dynamic cart system**, enabling customers to seamlessly add, remove, and modify items in real time with improved usability
- Built the frontend with **React JS** to deliver a responsive and user-friendly experience, and implemented backend functionality in **Node.js** to handle **order management**, routing, and data flow between client and server
- Gained practical experience in **modular development practices**, efficient **state management** using **React** hooks, and robust **client-server communication**, strengthening overall understanding of scalable e-commerce systems

Cricket Scoring System | *Course Project* (March - April 2025)

Guide: Prof. Kameswari Chebrolu - Software Systems Lab

- Developed a real-time **Cricket Scoring Web Application** using **HTML**, **CSS**, **JavaScript**, and **Python**
- Implemented modular features for player management, score tracking, strike rotation, and over summaries with robust error handling, and used **Selenium** to automate webpage simulation for testing and input automation
- Integrated interactive **modals** for player input, live updates, and user-friendly match simulation experience

TECHNICAL SKILLS

Languages	C/C++, Python, Bash, Assembly, Verilog, HTML, CSS, JavaScript, SQL, R, Java
Software & Tools	Git, GitHub, Docker, Make, L ^A T _E X, Markdown, Sed, Awk, Jupyter, ReactJS, Fusion360
Crypto/Security	PyCryptodome, Hash Functions, Side-Channel Analysis, Constant-Time Programming
Data/ML Libraries	NumPy, Pandas, Matplotlib, SciPy, TensorFlow, PyTorch, OpenCV, Scikit-learn
Embedded Systems	Arduino, FPGA (Verilog), Circuit Design, PCB Design, IMU/Gyroscope Sensors